

22 Probability Density Functions of Continuous Random Variables

- The continuous analog of a probability mass function (pmf) is a *probability density function (pdf)*.
- However, while pmfs and pdfs play analogous roles, they are different in one fundamental way; namely, a pmf outputs probabilities directly, while a pdf does not.
- For a continuous random variable X with pdf f_X , the probability that X takes a value in the interval $[a, b]$ is the *area under the pdf over the region $[a, b]$* .
- The **probability density function (pdf)** (a.k.a. density) of a *continuous* RV X is the function f_X which satisfies

$$P(a \leq X \leq b) = \int_a^b f_X(x)dx, \quad \text{for any } -\infty \leq a \leq b \leq \infty \quad (22.1)$$

- The axioms of probability imply that a valid pdf must satisfy

$$f_X(x) \geq 0 \quad \text{for all } x, \quad (22.2)$$

$$\int_{-\infty}^{\infty} f_X(x)dx = 1 \quad (22.3)$$

- A pdf is 0 outside the range of possible values. Given a specific pdf, the generic bounds like $(-\infty, \infty)$ in integrals should be replaced by the range of possible values.

Example 22.1. Recall Example 21.3. Based on earlier examples, it seems reasonable to assume that the pdf f_X is linear as a function of possible x values, with an intercept of 0 and positive slope; that is,

$$f_X(x) = \begin{cases} cx, & 0 < x < 60, \\ 0 & \text{otherwise.} \end{cases}$$

1. Compute the value of c . Note: this involves more than just reading off a value from a plot. What is the key principle that will determine the value of c ?

2. Use the pdf (and geometry) to compute $P(X < 15)$.
3. Use the pdf (and geometry) to compute $P(X > 45)$.
4. Use the pdf (and geometry) to find an expression for the cdf of X . (Compare to the cdf from Example 21.3.)

- If X is a continuous random variable with pdf f_X , its cdf F_X satisfies

$$F_X(x) = \int_{-\infty}^x f_X(u) du$$

- Roughly, the cdf is the integral of the pdf

Example 22.2. Continuing Example 22.1. Han's arrival time X (minutes after noon) has pdf

$$f_X(x) = \begin{cases} (2/3600)x, & 0 < x < 60, \\ 0 & \text{otherwise.} \end{cases}$$

The corresponding cdf satisfies $F_X(x) = (x/60)^2, 0 < x < 60$.

1. Compute and interpret $P(X = 15)$ and $P(X = 45)$.
2. Compute the probability that X , truncated to the nearest minute, is 15 minutes after noon; that is, compute $P(15 \leq X < 16)$. How does this probability compare to $f(15)$?

3. Compute the probability that X , truncated to the nearest minute, is 45 minutes after noon; that is, compute $P(45 \leq X < 46)$. How does this probability compare to $f(45)$?
4. Compute and interpret the ratio of the probabilities from the two previous parts. How does this ratio compare to $f(45)/f(15)$?

- The probability that a continuous random variable X equals any particular value is 0. That is, if X is continuous then $P(X = x) = 0$ for all x .
- In practical applications involving continuous random variables, “equal to” really means “close to”, and “close to” probabilities correspond to intervals which can have positive probability.
- The density $f_X(x)$ at value x is *not* a probability. Rather the height of the pdf $f_X(x)$ is related to $P(X \text{ is close to } x)$ in the following sense.
- Let ϵ be a small number in the measurement units of X . Then

$$P(x - 0.5\epsilon < X < x + 0.5\epsilon) \approx f_X(x)\epsilon$$

For example, $\epsilon = 0.01$ represents “close to” as rounding to two decimal places.

- In other words,

$$f_X(x) \approx \frac{P(x - 0.5\epsilon < X < x + 0.5\epsilon)}{\epsilon} = \frac{F_X(x + 0.5\epsilon) - F_X(x - 0.5\epsilon)}{\epsilon}$$

- Taking the limit as $\epsilon \rightarrow 0$, we see that the pdf is the derivative of the cdf

$$f_X(x) = \frac{d}{dx}F_X(x)$$

- For any two possible values x_1 and x_2

$$\frac{P(X \text{ is close to } x_1)}{P(X \text{ is close to } x_2)} \approx \frac{f_X(x_1)}{f_X(x_2)}$$

regardless of how “close to” is defined (as long as it’s reasonably “small”)

Example 22.3. Continuing Example 21.2. Let X be the times (minutes) until the next earthquake occurs (of any magnitude in a certain region). Assume that the pdf of X is

$$f_X(x) = (1/120)e^{-x/120}, \quad x \geq 0$$

1. Sketch the pdf of X . What does this tell you about waiting times?
2. Without doing any integration, approximate the probability that X rounded to the nearest second is 30 minutes.
3. Without doing any integration determine how much more likely that X rounded to the nearest second is to be 30 than 60 minutes.
4. Compute and interpret $P(X < 30)$.
5. Compute and interpret $P(X > 360)$.
6. Find the cdf of X .

7. Find the 25th percentile of X .
8. Find the quantile function of X .
9. Sketch a spinner for simulating values of X .